Hierarchical Context Merging:

Better Long Context Understanding for Pre-trained LLMs

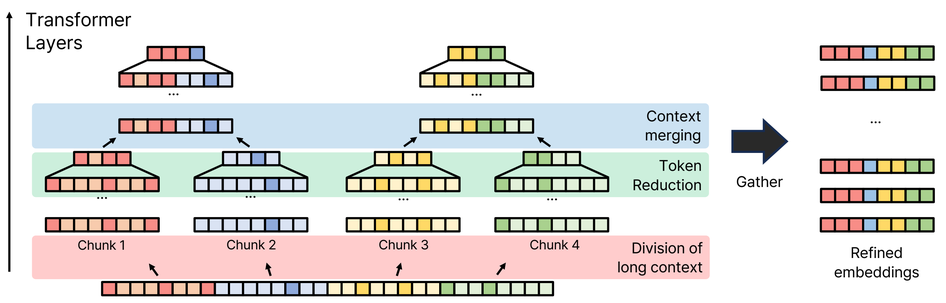
Woomin Song*, Seunghyuk Oh*,
Sangwoo Mo, Jaehyung Kim, Sukmin Yun, Jung-Woo Ha, Jinwoo Shin

HOMER
LONGER x16 context length
FASTER 160% faster generation
CHEAPER -70% peak memory
EASIER All without training

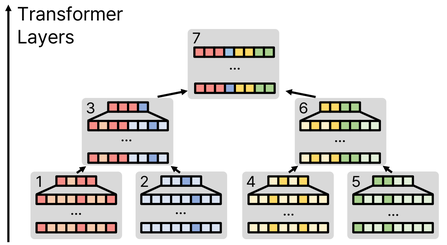
Just D'oh it!

HOMER (Hierarchical cOntext MERging)

HOMER
LONGER
FASTER
CHEAPER
EASIER



Hierarchical merging of context embeddings | **Propagative refinement**



Computation schedule

Experimental Results

Performance on Downstream Tasks

Pass key retrieval

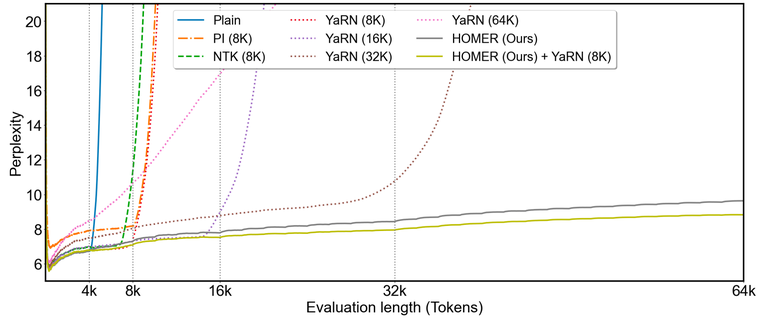
Method	Context limit	*Llama-2-7b-chat* 4K	8K	16K	32K	*Llama-2-13b-chat* 4K	8K	16K	32K
Plain	4k	**1.000**	0.000	-	-	**1.000**	0.000	0.000	0.000
Plain + HOMER	None	0.990	**0.924**	**0.890**	**0.776**	**1.000**	**0.944**	**0.882**	**0.804**
PI	8k	0.432	0.356	0.000	-	0.600	0.544	0.000	0.000
	16k	0.006	0.006	0.006	0.000	0.022	0.028	0.018	0.000
NTK	8k	0.812	0.000	0.000	0.000	0.866	0.000	0.000	0.000
	16k	0.516	0.652	0.000	0.000	0.626	0.692	0.000	0.000
	32k	0.106	0.194	0.162	0.000	0.286	0.570	0.442	0.000
YaRN	8k	**0.996**	0.002	0.000	-	**1.000**	0.464	0.000	0.000
	16k	0.844	0.756	0.000	0.000	0.980	0.952	0.214	0.000
	32k	0.702	0.654	0.696	0.002	0.926	0.888	0.836	0.026
	64k	0.678	0.358	0.148	0.026	0.902	0.826	0.364	0.224
YaRN + HOMER	None	**0.996**	**0.984**	**0.876**	**0.802**	**1.000**	**1.000**	**0.974**	**0.860**

Question Answering

Method	Context limit	Accuracy
Plain	4k	0.327
Plain + HOMER	None	**0.358**
PI	8k	0.366
NTK	8k	0.379
YaRN	8k	0.310
NTK + HOMER	None	**0.388**

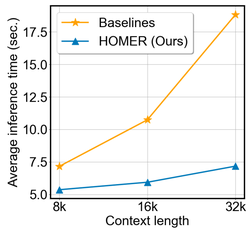
Fluency: Language Modeling Experiments

- Maintain low perplexity (↓) values up to 64k tokens (x16 original limit)

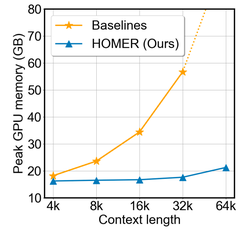


Efficiency of HOMER

- It saves up to 70% memory with a 160% speed-up



Inference Time



Memory

Setup	Average run time (seconds) 8k	16k	32k
	20 tokens		
Baselines*	1.879	3.224	6.546
HOMER	1.673 (12.3% speedup)	2.270 (42.0% speedup)	3.513 (86.3% speedup)
	50 tokens		
Baselines*	3.842	6.028	11.143
HOMER	3.026 (27.0% speedup)	3.639 (65.6% speedup)	4.873 (128.7% speedup)
	100 tokens		
Baselines*	7.149	10.733	18.828
HOMER	5.355 (33.5% speedup)	5.930 (81.0% speedup)	7.169 (162.6% speedup)

Setup	Peak GPU memory (GB) 4k	8k	16k	32k	64k
Baselines	18.2	23.6	34.4	56.7	> 80
HOMER	16.3 (-10.8%)	16.5 (-30.1%)	16.7 (-51.4%)	17.6 (-68.9%)	21.3 (at least -73.4%)
+ Baselines	19.2 (+5.5%)	20.1 (-14.6%)	20.3 (-41.1%)	20.8 (-63.4%)	22.5 (at least -71.8%)